清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

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Assessment of Iron and Steel Technologies from A Global Perspective

Tsinghua University & Saudi Aramco

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清华大学气候变化与可持续发展研究院
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Project Timeline

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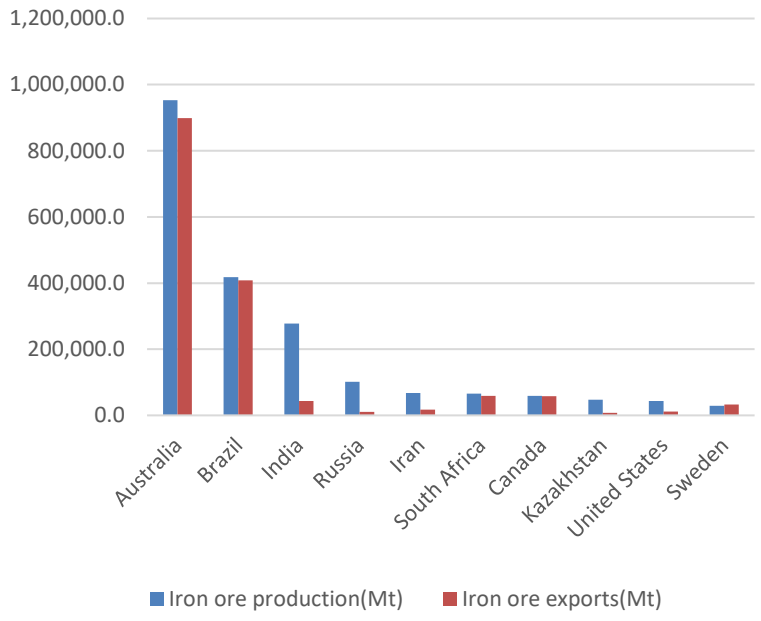
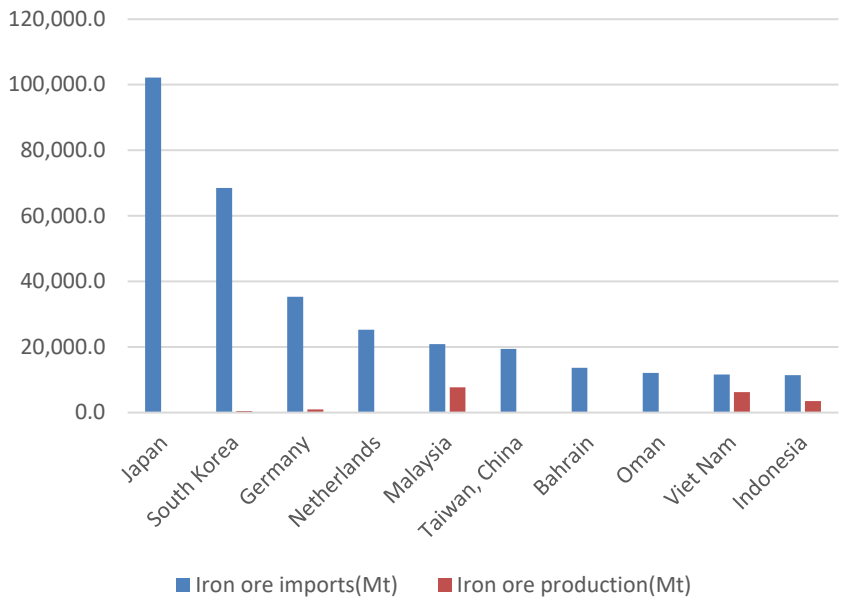
	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July
Research Proposal & preparations																								
Overview of global steel trade & technology assessment																								
Data requirement & database investigation																								
Model development																								
Case study: India																								
Case study: USA & Saudi Arabia																								
Case study: Australia & Japan																								
Case study: Germany & Russia																								
Case study: China-Australia steel trade route																								
Final report																								

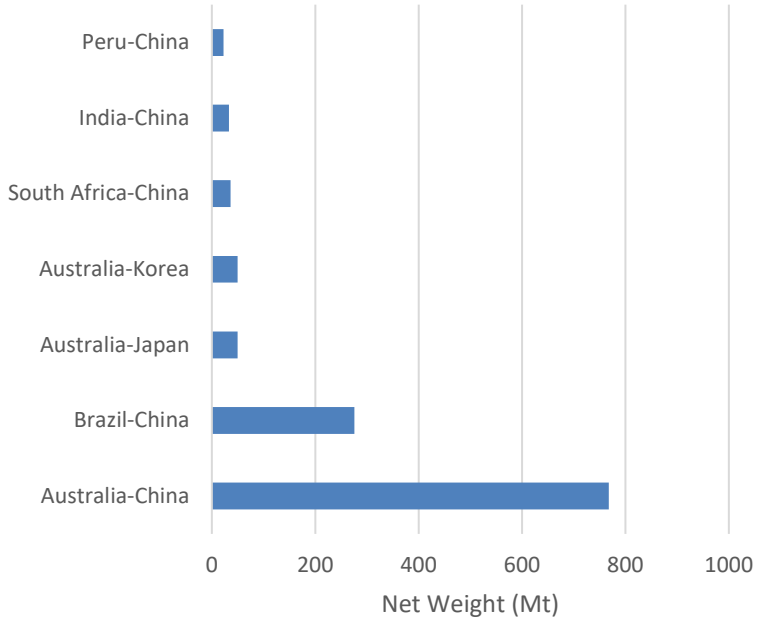
Global Steel Trade: Background



Iron ore production & exports in selected countries (Mt, 2023, Top10 iron ore producing countries)



Iron ore import & production in selected countries (Mt, 2023, Top10 iron ore import countries)

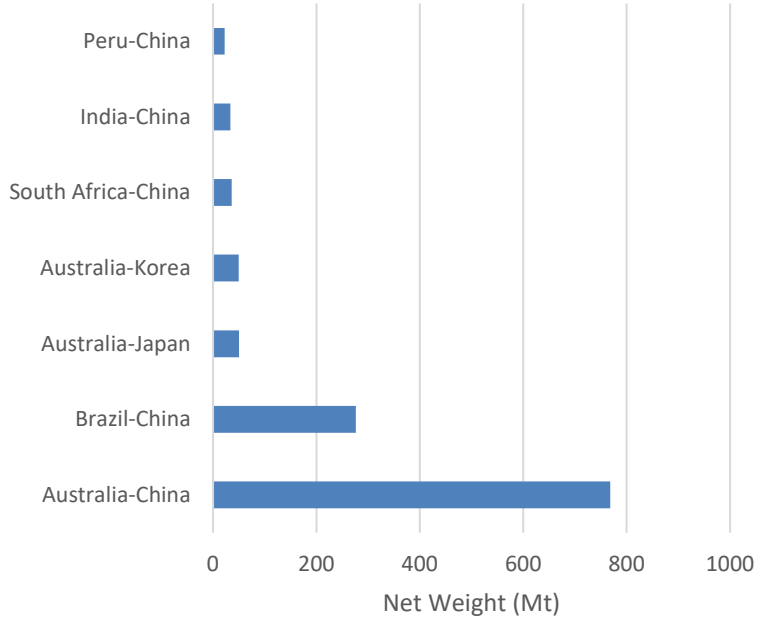


The world's largest iron ore trade routes (by volume)

- The steel industry is highly internationalized, with iron ore trade being a key component.
- The iron ore trade routes with the highest traffic volumes often emerge between major steel-producing regions and East Asian countries.
- There are countries where the mining industry depends on iron ore exports, or the steel industry depends on iron ore imports.

Global Steel Trade: Background

Overview of Indian Steel Industry



The world's largest iron ore trade routes (by volume)

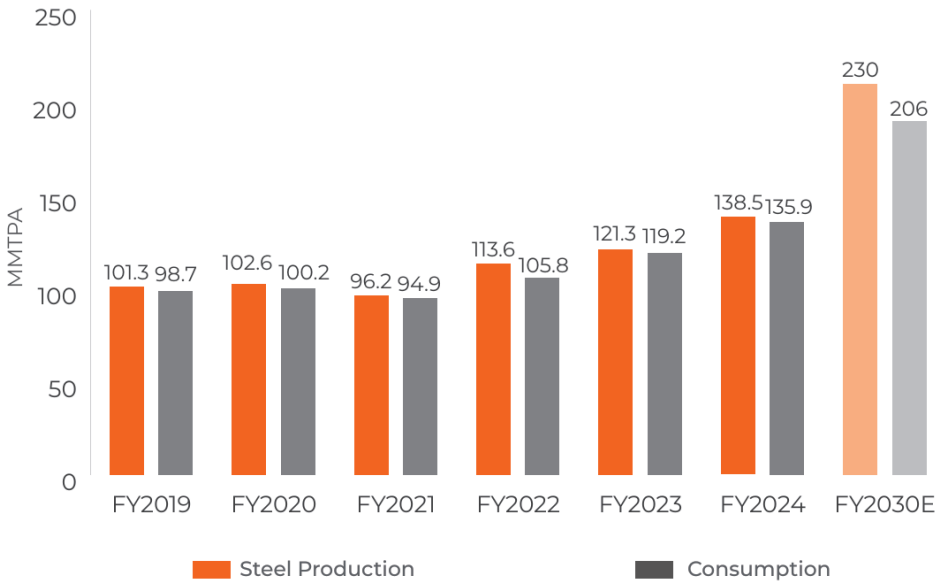
Classification by Global Steel Trade Status

Global Steel Trade Status	Description	Representative Countries
Mining	Mines iron ore, mostly using for export	Australia
Smelting	Relies on imports to gain iron ore, then smelt them to steel products	China, Japan, Saudi Arabia
Mining and Smelting	Relies on domestic mining to obtain iron ore, then smelt them to steel products	Russia, USA, India

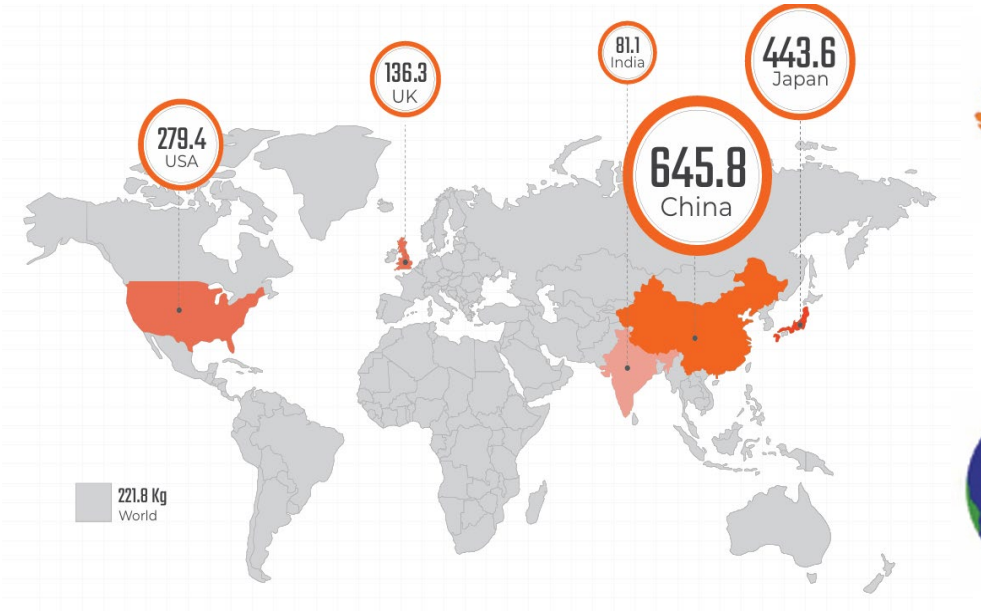
- Most of the major countries in the steel market can be classified into two categories: mining and smelting.
- The iron ore trade routes with the highest traffic volumes often emerge between mining countries and smelting countries.
- A small number of countries, leveraging their own abundant iron ore resources and relatively low costs, fall into a combined "mining & smelting" category.

India Case: Background

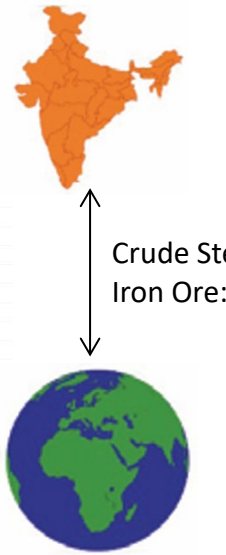
Overview of Indian Steel Industry



India's steel production and consumption volume in recent years



Comparison of Annual Per Capita Steel Consumption: India vs. Key Countries



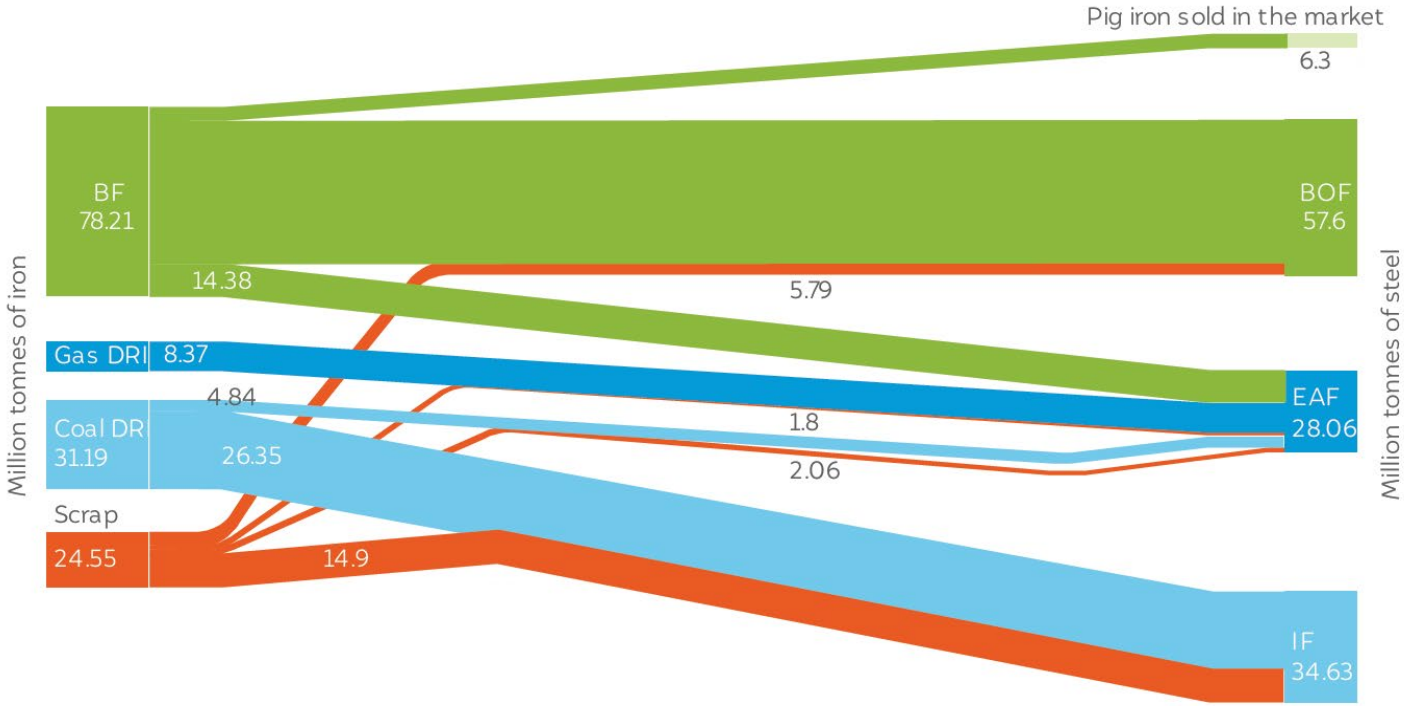
Crude Steel: Net Import of 2.3Mt
Iron Ore: Net Export of 14 Mt

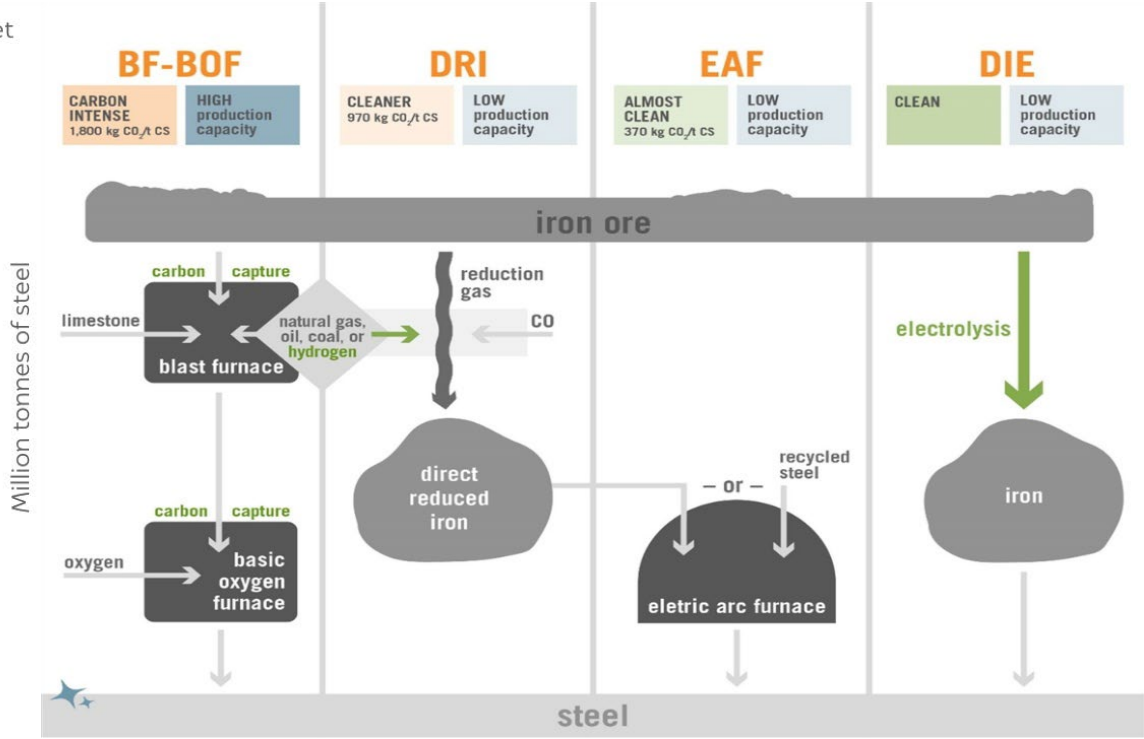
Import & output for Indian steel industry

- India's steel consumption & production's growth is poised to continue, driven by strong potential for an increase in per capita steel consumption, alongside immense demand from infrastructure development, urbanization, and the manufacturing sector.
- The vast majority of India's steel demand is met by its domestic manufacturing industry.
- India's iron ore production is sufficient to cover the consumption needs.

India Case: Background

Main technical routes of the India steel industry



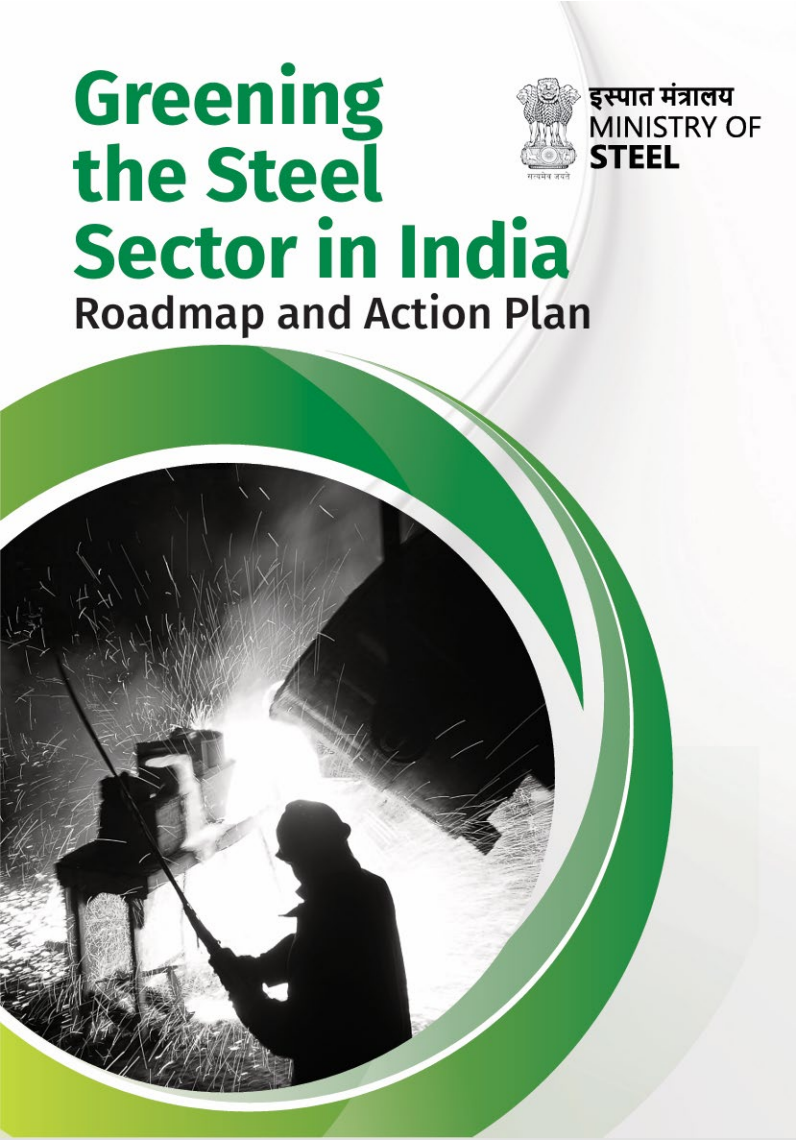


Technology process	reductant	CO2 emission intensity (tCO_2/tcs, India)
BF-BOF	Coke	**2.46**
DRI-IF	Coal	**2.30**
DRI-EAF	Coal	**2.51**
	NG	**1.91**

- The Coal-DRI is a unique coal-based Direct Reduced Iron (DRI) route specific to India.
- In India, the production scale of BOF (Basic Oxygen Furnace) steelmaking is roughly equivalent to that of EAF/IF steelmaking.
- The carbon emission intensity of the Coal-DRI process in India is higher than traditional BF-BOF route.

India Case: Background

Carbon intensity and decarbonization policies



Timeline	Objective
2030	• Reduce integrated emission intensity to 2.20 tCO_2/tcs • 45% penetration rate of renewable energy • Significantly increase the utilization of scrap steel and pellet ore • Demonstrate pilot plants using green hydrogen
2070	• Achieve net-zero emissions for the steel industry

- India emits 2.36 to 2.6 tonnes of carbon dioxide per tonne of crude steel produced, compared to the global average of 1.85 to 1.91 tCO_2/tcs
- The steel industry contributes approximately 2% to India's GDP, but accounts for 12% of the country's carbon emissions
- India has formulated an ambitious decarbonization policy plan, and emphasize new technologies like green hydrogen steelmaking

Thank You For Your Attention!